STAC AI BOOTCAMP NYC



Post-training for financial applications

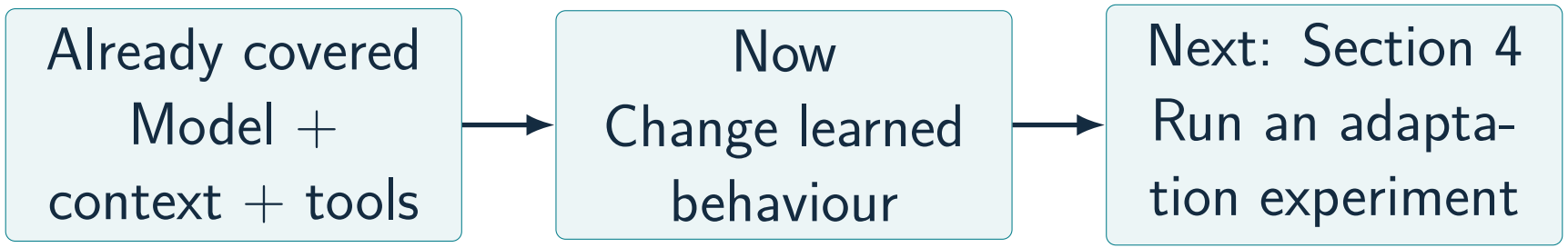
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What changes in Section 3?

Notebook: `03_post_training.ipynb`



Why train? Better information does not automatically teach a model how to respond.

Post-training means additional training of an existing model.

https://colab.research.google.com/github/mfrdixon/STAC_AI_Bootcamp/blob/main/notebooks/03_post_training.ipynb

Five things to be able to explain

- What an SFT demonstration teaches and which tokens are scored.
- Why LoRA is a parameter-update method, not a training objective.
- What rank and quantization change.
- How DPO and conventional RLHF use preference feedback.
- What independent evaluation must establish.

Start with three separate questions

Question	Options	What it controls
What should improve?	SFT; preference learning	The learning signal.
Which parameters can change?	Full tuning; LoRA	The space of trainable updates.
How is the model stored?	Higher precision; quantized	Memory and numerical representation.

These choices combine: SFT + LoRA; DPO + LoRA; SFT + QLoRA.

SFT: supervised fine-tuning — 1 of 3

1. Provide a reviewed demonstration	A record contains the task, input and response we want the model to learn. The target response is the label.

In plain English: Example: input The meeting is on Tuesday; reviewed completion Tuesday.

SFT: supervised fine-tuning — 2 of 3

1. Provide a reviewed demonstration	A record contains the task, input and response we want the model to learn. The target response is the label.
2. Reward the recorded answer tokens	Training increases their conditional likelihood. It does not automatically check whether the reviewer answer is true.

In plain English: Example: input The meeting is on Tuesday; reviewed completion Tuesday.

SFT: supervised fine-tuning — 3 of 3

1. Provide a reviewed demonstration	A record contains the task, input and response we want the model to learn. The target response is the label.
2. Reward the recorded answer tokens	Training increases their conditional likelihood. It does not automatically check whether the reviewer answer is true.
3. Test on different cases	A model can fit the demonstrations without handling new wording or completing the whole workflow correctly.

In plain English: Example: input The meeting is on Tuesday; reviewed completion Tuesday.

What one training record contains

Part	Example	Purpose
Instruction	Extract the meeting day.	Defines the task.
Input	The meeting is on Tuesday.	Provides the context.
Target response	Tuesday.	Provides the demonstration.

The chat template converts these parts into one token sequence.

Decide explicitly: which response positions contribute to loss?

Teacher forcing: predict each next target token



Training supplies the correct preceding answer tokens.

At generation time, the model instead continues from its own previous output.

A small labelled SFT dataset

Input record	Reviewed target label
Extract the day: The meeting is on Tuesday.	Tuesday.
Extract the current day: The meeting moved from Tuesday to Friday.	Friday.

The response is the label. In a prompt–completion dataset it is stored as the completion.

The examples teach a task, not a table of confidence scores.

The loss mask selects what is scored — 1 of 3

1. Keep the question visible	The model still reads the instruction and input while predicting the response.

In plain English: Changing a zero-loss prompt can still change the scored answer predictions.

The loss mask selects what is scored — 2 of 3

1. Keep the question visible	The model still reads the instruction and input while predicting the response.
2. Score the selected answer positions	A mask of one includes a prediction error. A mask of zero excludes that position from direct loss.

In plain English: Changing a zero-loss prompt can still change the scored answer predictions.

The loss mask selects what is scored — 3 of 3

1. Keep the question visible	The model still reads the instruction and input while predicting the response.
2. Score the selected answer positions	A mask of one includes a prediction error. A mask of zero excludes that position from direct loss.
3. Do not confuse this with attention	A loss mask selects errors to score. An attention mask controls which positions can interact.

In plain English: Changing a zero-loss prompt can still change the scored answer predictions.

Before the equation: what is assumed?

- The model predicts each token conditional on the preceding tokens; tokens are not assumed independent.
- Reviewed targets represent the behavior to imitate. Wrong labels teach the wrong behavior.
- This example uses teacher forcing and response-only scoring.
- The displayed reduction is a token-weighted average; other weighting conventions are possible.

SFT loss: one average over scored tokens

$$\mathcal{L}_{\mathrm{SFT}}(\theta) = -\frac{1}{M} \sum_{i=1}^{N} \sum_{t=1}^{T_i} m_{i,t} \log p_\theta(z_{i,t} \mid z_{i,<t})$$

- θ: trained parameters; $\mathcal{L}_{\mathrm{SFT}}$: mean supervised loss; N: examples; i: example index; T_i: token count; t: token position.
- $z_{i,t}$: current token; $z_{i,<t}$: preceding tokens; p_θ: model probability; $|$: conditional on.
- $m_{i,t}$: mask, 1 if scored and 0 otherwise; $M = \Sigma_{i,t}\, m_{i,t} > 0$: total scored tokens; Σ: sum; log: natural logarithm.
- Read: average negative log probability of the demonstrated tokens. Loss is in nats per scored token; prompt masking does not remove context.

A tiny calculation makes the loss concrete

Scored token	Assigned probability	Negative log probability
First target token	0.8	0.223
Second target token	0.5	0.693

$$\text{Mean loss} = (0.223 + 0.693)/2 = 0.458$$

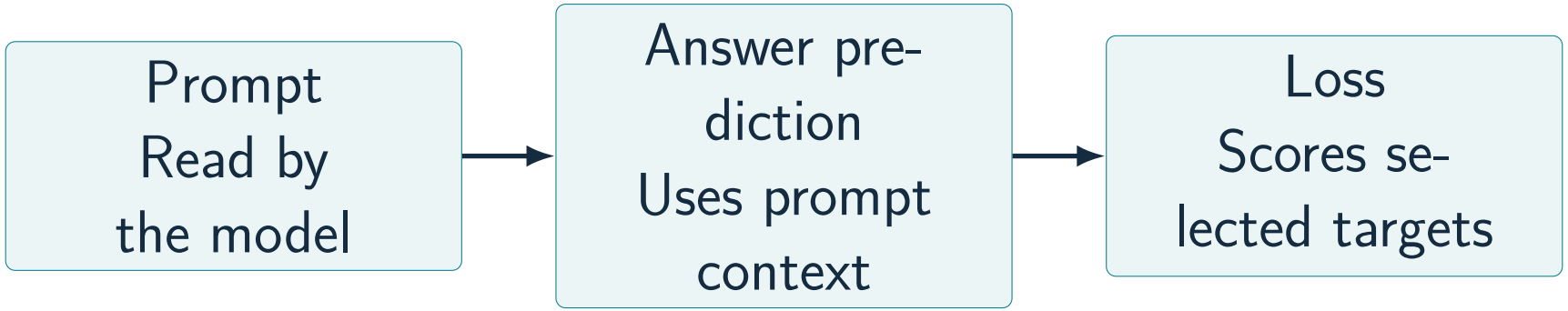
Higher probability on recorded targets lowers loss.

It does not establish success on unseen tasks.

Check 1: what does a zero loss mask mean?

- A. The model cannot read that position.
- B. That position has no direct contribution to the loss.
- C. The model assigns that token zero probability.

Answer: B — excluded from direct loss



The prompt remains available to help predict the response.

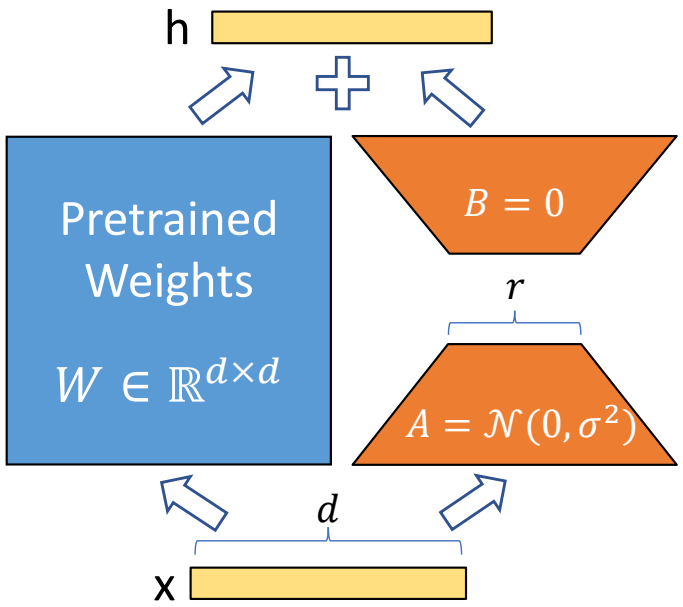
The loss mask only selects which prediction terms enter the objective.

Why not update every parameter?

- A large model contains billions of weights. Updating all of them needs gradients and optimizer memory.
- A separate full model for each task is also expensive to store and maintain.
- LoRA trains a small correction while keeping the original weights fixed.

Purpose: make task adaptation feasible with fewer trainable parameters.

LoRA: low-rank adaptation



Read the original figure

Blue: the original transformation stays fixed.

Orange: small matrices A and B learn a correction.

Both paths receive x; their outputs are added to produce h.

The added branch is inside a selected layer, not a separate model between layers.

x: input; h: output; W_0: frozen matrix; A, B: trainable correction matrices; r: positive rank; α: scale. The multiplier α/r is unitless.

LoRA equation: base transformation + correction

$$h = W_0 x + \frac{\alpha}{r} BAx$$

- x: input vector; h: output vector; W_0: frozen base matrix, with output-by-input shape.
- A: trainable input-to-rank matrix; B: trainable rank-to-output matrix; r: positive integer rank; α: scale; α/r: unitless multiplier.
- Read: add a learned low-rank correction to the same base transformation. Vector units depend on the representation; weights map between them.
- Only the selected adapter matrices are updated. This is a parallel branch inside a layer, not a smaller replacement model or a layer inserted between every pair of layers.

What changes when rank increases?

Rank for a 4096 by 4096 matrix	Added trainable parameters	Interpretation
4	32,768	Smaller update space.
8	65,536	Twice as many trainable entries.
16	131,072	More update capacity and resource use.

Larger rank permits a richer correction; it does not guarantee a better model.

Choose rank using held-out evaluation and the compute budget.

Who chooses where LoRA is applied?

Choice	Who determines it?
Target modules: Q, K, V, output or feed-forward projections	Developer configuration.
Rank and scaling	Developer configuration, evaluated experimentally.
Values inside A and B	Backpropagation and the optimizer.

```
LoraConfig(r=8, target_modules=["q_proj", "v_proj"])
```

Q/V is a choice, not a universal optimum. Compare alternatives on the task.

Why QLoRA? LoRA still keeps a large base in memory

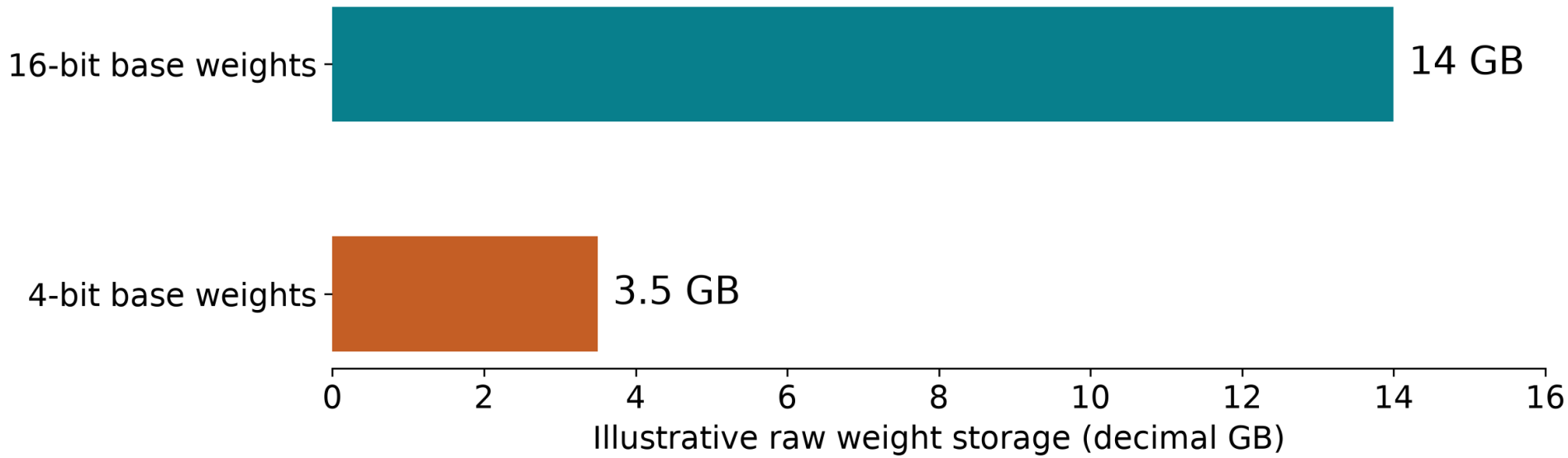


Illustration: 7 billion weights $\times$ 16 bits $=$ 14 GB; at 4 bits $=$ 3.5 GB.

Raw weights only: excludes scales, adapters, activations and runtime overhead.

What does storing a weight in 4 bits mean? — 1 of 3

1. A large frozen base still occupies memory	LoRA reduces what is trained, but the original base weights remain needed for computation.

In plain English: Smaller memory requirements may enable a smaller GPU or a longer input; total cost savings are workload-dependent.

What does storing a weight in 4 bits mean? — 2 of 3

1. A large frozen base still occupies memory	LoRA reduces what is trained, but the original base weights remain needed for computation.
2. Four bits encode sixteen codes	Codes select approximate numerical levels, with scaling information. This compact representation saves base-weight storage.

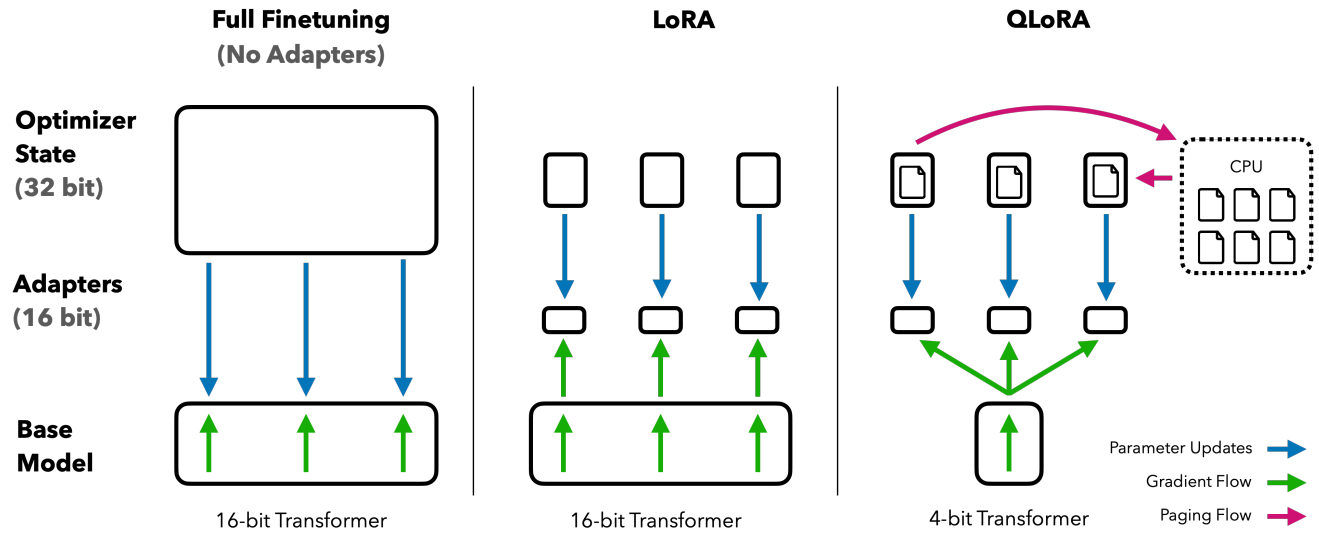
In plain English: Smaller memory requirements may enable a smaller GPU or a longer input; total cost savings are workload-dependent.

What does storing a weight in 4 bits mean? — 3 of 3

1. A large frozen base still occupies memory	LoRA reduces what is trained, but the original base weights remain needed for computation.
2. Four bits encode sixteen codes	Codes select approximate numerical levels, with scaling information. This compact representation saves base-weight storage.
3. Storage precision is not all computation	Adapters and calculations can use higher precision. Quantization adds approximation and does not guarantee the same quality or speed.

In plain English: Smaller memory requirements may enable a smaller GPU or a longer input; total cost savings are workload-dependent.

QLoRA: LoRA with a quantized base



LoRA: fewer trainable parameters. **QLoRA:** compact frozen base too.

Smaller memory requirements can enable a smaller GPU or fewer GPUs; speed and cost savings depend on the workload.

Check 2: can LoRA be used with SFT and DPO?

- A. No. LoRA replaces both training objectives.
- B. Only SFT; LoRA requires one target response.
- C. Yes. Either objective can train LoRA parameters.

Answer: C — objective and update method combine

Learning signal	Trainable parameters	Valid combination
SFT demonstrations	Full model or LoRA	SFT + LoRA.
DPO preference pairs	Full model or LoRA	DPO + LoRA.
Either, with quantized frozen base	LoRA adapters	QLoRA-style configuration.

SFT/DPO: what the loss rewards.
LoRA: how the parameter change is represented.

Why move beyond demonstrations?

- Writing one ideal answer can be difficult or expensive.
- A reviewer may find it easier to compare two existing answers.
- Preference data captures which response is better under a stated criterion.

Example: compare two summaries for factual support and clarity, rather than ask the reviewer to write a new summary.

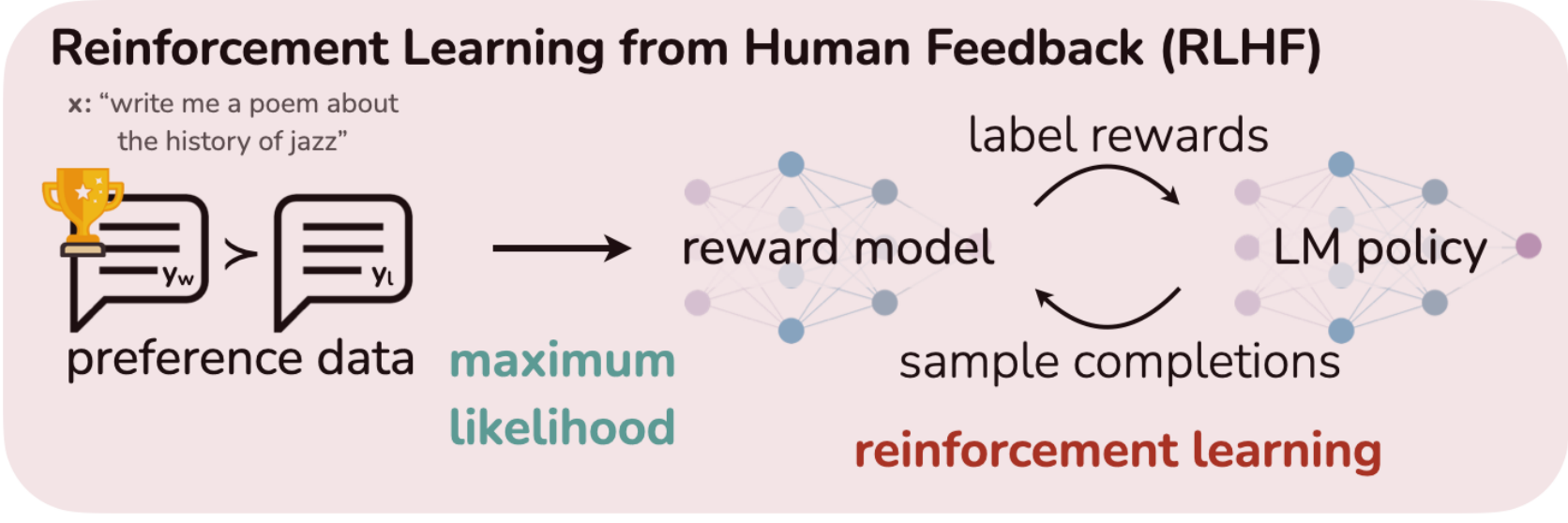
RLHF: the goal; PPO and DPO: different routes

Goal: use human feedback to improve responses.

Conventional reward-model + PPO route	DPO route
Learn a reward model, then optimize responses against its scores.	Train directly on recorded chosen/rejected response pairs.
Includes an online generation-and-scoring loop during policy training.	Standard offline DPO does not require that loop or a separately trained reward model.

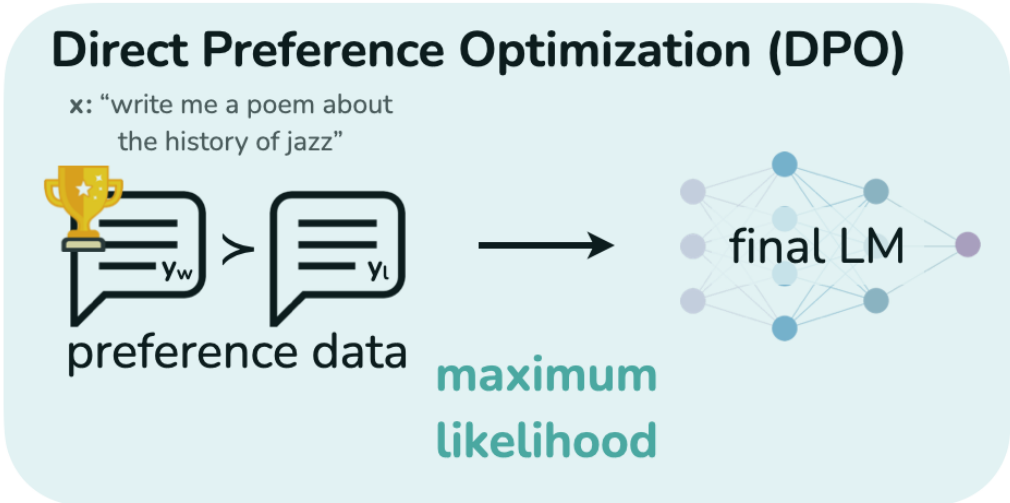
DPO is an alternative preference-training recipe, not the opposite of learning from human feedback.

RLHF: reinforcement learning from human feedback



Two jobs: learn a reward predictor; then use it to score generated responses and update the language model.

DPO removes the reward-model training loop



One training path: recorded preference pairs directly update the language model.

No separate reward model or on-policy generation loop in standard offline DPO.

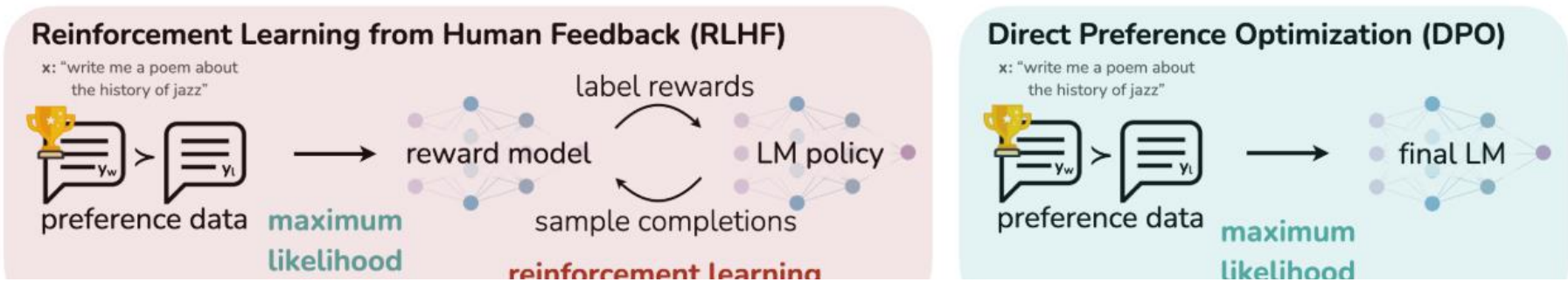
The original lecture makes the same comparison

Try to simplify PPO by..

- Getting rid of the reward model
- Getting rid of any on-policy stuff (rollouts, outer loops etc)

Instead

- Take gradient steps on log-loss of good stuff
- Take negative gradient steps on bad stuff (appropriately weighted).



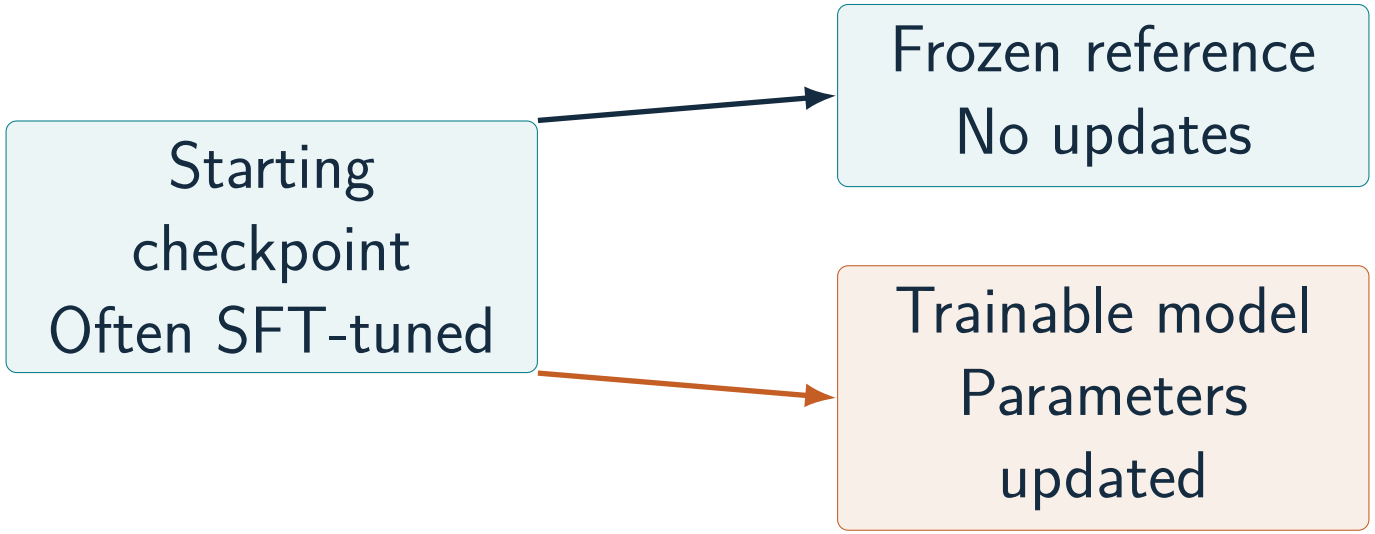
A DPO record: the preference label is explicit

Prompt and evidence	Chosen	Rejected
Meeting moved from Tuesday to Friday. Which day now?	Friday.	Tuesday.
No meeting day is provided. Which day?	The day is not specified.	Tuesday.

Stored fields: `prompt`, `chosen`, `rejected`.

The label is a preference, not an independently verified truth probability.

The reference model is a frozen starting point



Both score the same chosen and rejected responses.

Two meanings of frozen: DPO reference copy versus LoRA base weights.

The reference model is a frozen starting point — 1 of 3

1. Begin with a useful checkpoint	Often this is an already instruction-following model. A new local SFT stage is common but not always required.

In plain English: Freezing the DPO reference is a different role from freezing base weights while training LoRA adapters.

The reference model is a frozen starting point — 2 of 3

1. Begin with a useful checkpoint	Often this is an already instruction-following model. A new local SFT stage is common but not always required.
2. Keep one reference fixed	It scores the same candidate answers without updating. It is not the reviewer, dataset or source of factual truth.

In plain English: Freezing the DPO reference is a different role from freezing base weights while training LoRA adapters.

The reference model is a frozen starting point — 3 of 3

1. Begin with a useful checkpoint	Often this is an already instruction-following model. A new local SFT stage is common but not always required.
2. Keep one reference fixed	It scores the same candidate answers without updating. It is not the reviewer, dataset or source of factual truth.
3. Update the response model	DPO measures how the chosen-versus-rejected likelihood ratio changes relative to that reference.

In plain English: Freezing the DPO reference is a different role from freezing base weights while training LoRA adapters.

DPO rewards a relative preference shift — 1 of 3

1. Compare two answers to the same prompt	The dataset identifies a chosen and a rejected response. A preference criterion explains why one is favored.

In plain English: Illustration: moving a chosen/rejected ratio from 1/2 to 2 is a fourfold relative shift, not a probability that the answer is true.

DPO rewards a relative preference shift — 2 of 3

1. Compare two answers to the same prompt	The dataset identifies a chosen and a rejected response. A preference criterion explains why one is favored.
2. Compare current and starting preferences	Both models score both responses. The loss rewards improving the chosen/rejected ratio relative to the fixed reference.

In plain English: Illustration: moving a chosen/rejected ratio from 1/2 to 2 is a fourfold relative shift, not a probability that the answer is true.

DPO rewards a relative preference shift — 3 of 3

1. Compare two answers to the same prompt	The dataset identifies a chosen and a rejected response. A preference criterion explains why one is favored.
2. Compare current and starting preferences	Both models score both responses. The loss rewards improving the chosen/rejected ratio relative to the fixed reference.
3. Do not interpret preference as truth	A lower DPO loss means better fit to preference labels. It does not establish factual accuracy, calibrated beliefs or better business outcomes.

In plain English: Illustration: moving a chosen/rejected ratio from 1/2 to 2 is a fourfold relative shift, not a probability that the answer is true.

DPO: connect the explanation to the equation

$$\Delta = \log \frac{\pi_\theta(y^+ \mid x)}{\pi_\theta(y^- \mid x)} - \log \frac{\pi_{\text{ref}}(y^+ \mid x)}{\pi_{\text{ref}}(y^- \mid x)}, \qquad \mathcal{L}_{\text{DPO}} = -\log \sigma(\beta\Delta)$$

- x: prompt; y^+, y^-: chosen/rejected responses; π_θ: trainable response distribution with parameters θ; π_{ref}: frozen reference.
- Δ: change in log preference ratio; log: natural logarithm; $|$: conditional on; $\beta > 0$: scaling parameter.
- $\sigma(u) = 1/(1 + \exp(-u))$: sigmoid for input u; exp: exponential; $\mathcal{L}_{\text{DPO}}$: per-pair DPO loss, unitless.
- Read: reward a stronger chosen-versus-rejected ratio relative to the reference. Requires paired preferences, positive candidate probabilities and fixed scoring conventions; not a calibrated truth probability.

What assumptions support the DPO equation?

- A prompt x has two candidate responses and one recorded preference; ties are omitted in this binary setup.
- A scalar latent reward and a logistic pairwise model describe preferences.
- The reference stays fixed and assigns nonzero probability to the scored responses.
- The standard derivation links this preference model to a KL-regularized reward objective.

Real preferences can violate these assumptions. Training can still fit the labels without proving their validity.

Why use DPO after SFT? — 1 of 3

1. SFT establishes a response pattern	Reviewed demonstrations can teach instruction following, output structure and a specific interpretation skill.

In plain English: Common sequence: learn a useful response style first, then refine a defined preference. This is not mandatory for every model.

Why use DPO after SFT? — 2 of 3

1. SFT establishes a response pattern	Reviewed demonstrations can teach instruction following, output structure and a specific interpretation skill.
2. DPO adjusts recurring preferences	Comparisons can favor supported, useful responses over plausible alternatives without requiring a new ideal answer every time.

In plain English: Common sequence: learn a useful response style first, then refine a defined preference. This is not mandatory for every model.

Why use DPO after SFT? — 3 of 3

1. SFT establishes a response pattern	Reviewed demonstrations can teach instruction following, output structure and a specific interpretation skill.
2. DPO adjusts recurring preferences	Comparisons can favor supported, useful responses over plausible alternatives without requiring a new ideal answer every time.
3. Do not add a stage without a need	SFT may already be sufficient. Add preference training only when reliable labels address a measurable recurring failure.

In plain English: Common sequence: learn a useful response style first, then refine a defined preference. This is not mandatory for every model.

SFT, DPO and conventional RLHF: what differs?

Method	Training information	Core training path
SFT	Prompt + demonstrated response.	Predict demonstrated tokens.
DPO	Prompt + chosen/rejected pair.	Optimize relative preference using a reference.
Conventional RLHF	Comparisons plus generated responses.	Train reward model; optimize policy.

All depend on data quality and evaluation. LoRA can reduce trainable parameters.

Do training records have to be i.i.d.?

- LoRA is a parameterization: it has no independent-record requirement.
- SFT and DPO losses can be computed on dependent records.
- Usual generalization arguments often assume independent or suitably controlled samples from a relevant task distribution.
- Repeated documents and near duplicates reduce effective information; keep related cases together when splitting.

Example: 100 paraphrases of one document are not 100 independent source cases.

Preference improvement is not probability calibration — 1 of 3

1. Preferred means favored under a rubric	A reviewer may prefer clarity or brevity. That preference is not automatically a check of every factual statement.

In plain English: Calibration needs resolved labels and separate evaluation; SFT or DPO alone provides no such guarantee.

Preference improvement is not probability calibration — 2 of 3

1. Preferred means favored under a rubric	A reviewer may prefer clarity or brevity. That preference is not automatically a check of every factual statement.
2. Correct means supported and accurate	A clear answer can still be false. A correct categorical answer does not validate its confidence number.

In plain English: Calibration needs resolved labels and separate evaluation; SFT or DPO alone provides no such guarantee.

Preference improvement is not probability calibration — 3 of 3

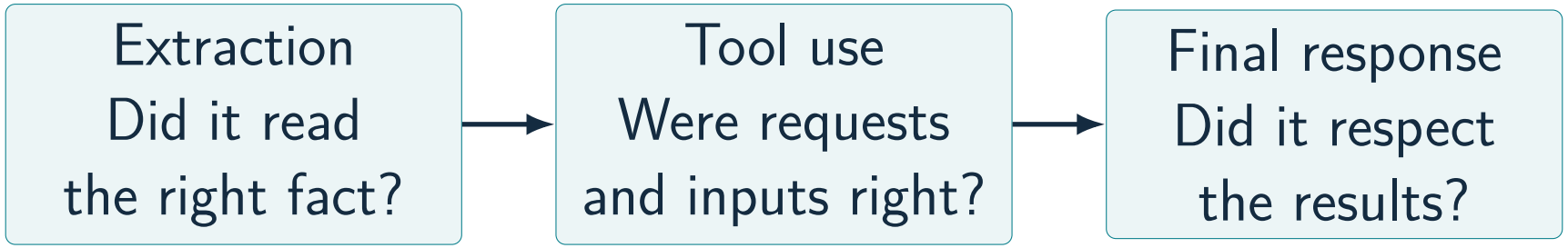
1. Preferred means favored under a rubric	A reviewer may prefer clarity or brevity. That preference is not automatically a check of every factual statement.
2. Correct means supported and accurate	A clear answer can still be false. A correct categorical answer does not validate its confidence number.
3. Calibrated means matching frequencies	For a defined event, cases scored near 80 percent should resolve positively about 80 percent of the time in the relevant population.

In plain English: Calibration needs resolved labels and separate evaluation; SFT or DPO alone provides no such guarantee.

Check 3: what would justify using the update?

- A. Lower loss on the training examples.
- B. More polished responses preferred by reviewers.
- C. Better held-out task performance with required checks preserved.

Answer: C — test behavior in the workflow



Compare base and adapted models on the same unseen cases.

A better average can conceal a new failure in a critical step.

Class exercise: identify what each setting changes

- Open `03_post_training.ipynb`; run setup C01.
- C02: change the masked prompt probability; inspect mean loss.
- C03: compare ranks, then trace input [2,1] to output [3,0].
- C04: compare chosen/rejected score pairs and DPO loss.
- These numerical examples explain the mechanism; Section 4 examines saved adaptation outcomes.

https://colab.research.google.com/github/mfrdixon/STAC_AI_Bootcamp/blob/main/notebooks/03_post_training.ipynb

Exercise review: three distinct answers

- Loss mask: selected response tokens are scored; prompt context remains visible.
- Rank: doubling the adapter rank doubles its trainable parameter count for fixed target matrices.
- Preference: the reviewer specifies which response is better and why.

https://colab.research.google.com/github/mfrdixon/STAC_AI_Bootcamp/blob/main/notebooks/03_post_training.ipynb

Choose the intervention that matches the problem

Observed problem	First intervention to investigate
Required information is absent.	Retrieval, source access and context construction.
Recurring task behavior remains wrong.	Instructions, demonstrations and SFT evaluation.
Responses violate a defined preference.	Reviewed comparisons; consider DPO.
An action exceeds delegated authority.	Enforced application permissions.

Training changes behavior; it does not replace evidence or authorization.

Objectives reviewed — next, run the experiment

Objective	What we have established
SFT	Demonstrations, teacher forcing and scored tokens.
LoRA / QLoRA	Trainable update, rank and base storage.
DPO / RLHF	Preference pairs, reference and reward pathways.
Evaluation	Independent cases and workflow checks.

Section 4: prepare data, run or inspect an adaptation, compare failures.

The question is whether the update improves the behavior we need.